

XR4DAvatar: Photorealistic 4D Avatar for Extended Reality

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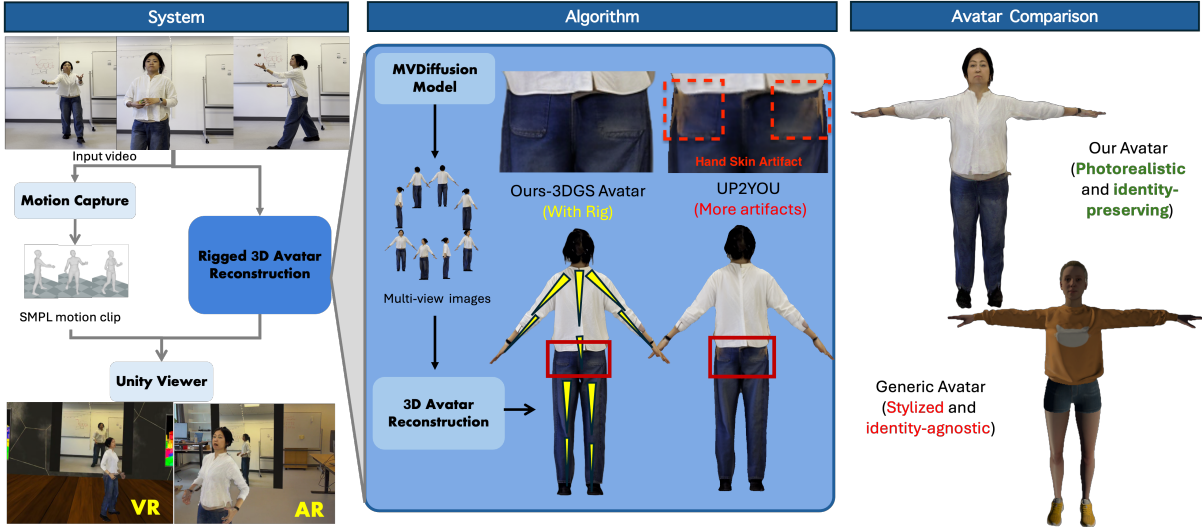


Figure 1: XR4DAvatar is a framework for reconstructing photorealistic 4D avatars from monocular in-the-wild videos. The left column illustrates the system pipeline, enabling rigged 3DGS avatar reconstruction and supporting direct animation and rendering in Unity for XR applications. The middle column presents an overview of our avatar reconstruction algorithm and a comparison with the baseline, showing that our method produces fewer artifacts and includes a rigged skeleton. The right column compares our avatar with a generic avatar, demonstrating higher realism and better identity preservation.

ABSTRACT

We present a novel framework for reconstructing photorealistic 4D avatars from monocular videos for XR applications. Unlike existing 3D Gaussian Splatting (3DGS)-based avatar reconstruction methods that rely on constrained human poses and camera setups and often lack animation rigs, our approach supports unconstrained real-world monocular videos as input and produces animatable, photorealistic avatars. We integrate a diffusion-based image generation model to generate multi-view images and a textured SMPL-X-guided 3D Gaussian Splatting model to construct 3D avatars. Our results show comparable visual quality to existing approaches while supporting direct animation. We further demonstrate a practical application of reconstructed avatars by directly driving 3DGS avatars with SMPL-based animation in Unity for both VR and AR.

Index Terms: Avatar Reconstruction, Extended Reality, Gaussian Splatting, Mixed Reality, Augmented Reality, Photorealism.

1 INTRODUCTION

Whole-body avatars play an essential role in various applications, such as surgical replay [2], collaborative meetings, and physical training [4]. As an interactive interface, avatars can convey complex non-rigid body movements, while their appearance can communicate non-verbal information and enhance social connection.

Traditional digital avatar creation requires appearance modeling and rigging, followed by motion capture and animation, which are labor-intensive and require specialized expertise and hardware. To reduce this cost, recent methods explore learning-based reconstruction from multi-view or monocular videos. In this work, we focus on a cost-efficient approach using only monocular in-the-wild videos. However, this setting remains challenging due to limited viewpoints. Existing inverse-rendering methods often overfit visible regions, producing artifacts and incomplete geometry on unseen body parts. As a result, it is non-trivial to achieve high-quality, complete, and animatable 3D avatars under monocular capture.

To address these challenges, several works integrate 3DGS methods with generative models to generate the unseen part or learn a generalized human appearance prior. However, most remain research-oriented and prioritize rendering quality over practical XR deployment. Dong et al. proposed a 3DGS-based avatar reconstruction and animation framework exportable to Unity [7]. To further reduce rendering overhead, Iandola et al. introduced SqueezeMe [3], which converts 3DGS into a lightweight representation. However, these methods assume either a static camera with the subject rotating in a canonical pose or multi-view images, limiting their use for in-the-wild videos. More recently, VRGaussianAvatar [6] enabled real-time 3DGS avatar rendering in VR from a single image, but streams rendered images directly to the HMD without supporting integration with customizable scene content.

To bridge this gap, we propose a 4D human performance replay framework from monocular videos for XR applications. For 3D avatar construction, we design a two-stage pipeline. Specifically, we first introduce a 2D diffusion-based image generation module to generate multi-view images of a human subject in a canonical pose. Based on these generated views, we then develop an SMPL-

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X-guided 3D Gaussian Splatting avatar reconstruction module for building a rigged, high-quality canonical avatar. To further improve the visual quality when the avatar is animated, a textured SMPL model is integrated as a base model, which also reduces the 3DGS numbers. Finally, we integrate an SMPL-X-based motion capture model for motion reconstruction together with a Unity-based rendering system for avatar rendering and animation.

In summary, our contributions are:

- A 3DGS avatar reconstruction method to support realistic avatar rendering and direct animation without manual rigging. Quantitative result shows comparable visual quality, while qualitative results show fewer artifacts.
- A Unity-based 3D avatar rendering and animation pipeline for VR and AR that bridges the gap between research-based avatar reconstruction methods and practical XR deployment. The pipeline is designed to support different 3DGS-based avatar reconstruction methods.

2 METHOD

3DGS avatar reconstruction Given a monocular video, we reconstruct a photorealistic and animatable avatar in A-pose. To mitigate incomplete observations, we use a diffusion model from UP2You [1] to generate multi-view images conditioned on A-pose normal maps and pose correlations. Based on the multi-view images, we construct a 3DGS avatar combined with a textured SMPL-X model. Unlike UP2You, which directly lifts RGB and normal cues into 3D without a rigged skeleton and exhibits noticeable artifacts, our method initializes Gaussian primitives from SMPL-X mesh vertices. Each Gaussian is parameterized by its offset relative to the SMPL-X surface, rotation, opacity, and color coefficients, and is optimized using a rendering-based photometric loss. To further improve compactness and rendering fidelity, we integrate a textured SMPL model as a structural prior. The final output is obtained by jointly rendering the 3DGS avatar and the textured SMPL mesh.

3DGS avatar animation To recover human motion from in-the-wild videos, we adopt the state-of-the-art motion estimation method GVHMR [5], which reconstructs world-grounded SMPL motion sequences. Given the recovered motion and the reconstructed 3DGS avatar, we build a 4D replay pipeline in Unity based on GSAC [7] to enable flexible XR deployment. Our framework bridges conventional animation pipelines with deformable 4D Gaussian Splatting rendering. The estimated SMPL motions are first converted into a Unity-compatible animation format via a custom motion conversion module. In Unity, an SMPL mesh is animated, and each Gaussian splat is updated using its stored offset relative to the associated SMPL surface. This allows global splat positions to follow mesh deformations during motion. Finally, 3D Gaussian splats are rendered as 2D splats and alpha-composited with the rasterized background, supporting real-time avatar animation in XR environments.

3 EXPERIMENTS AND DISCUSSION

All experiments were conducted on a single NVIDIA RTX 5080 GPU. For 4D replay, we developed a Unity project supporting both VR and AR. The system was implemented in Unity 6000.1.13f1, with the Meta Quest 3 serving as the target display platform.

For evaluating avatar reconstruction quality, we capture three real-world videos and front- and back-view images of subjects in an A-pose as reference following the evaluation method proposed in UP2YOU for in-the-wild videos. We compute perceptual similarity using CLIP and DINO-V2 [1] between rendered front and back views of the reconstructed avatars and the reference images. Our approach achieves higher perceptual similarity scores. In addition, because we use the same multi-view image generation in UP2YOU, to further evaluate the visual quality of the mesh-based method in

Table 1: Quantitative comparison

Method	SSIM $\uparrow$	PSNR $\uparrow$	CLIP-I $\uparrow$	DINO $\uparrow$
UP2You [1]	0.938	26.150	0.949	0.940
Ours	0.937	26.418	0.954	0.945

UP2YOU and our 3DGS-based method, we calculate SSIM and PSNR between front and back views of each reconstructed avatar and the generated multi-view images, which serve as pseudo ground truth, with results reported in Table 1. Our results show comparable visual quality to UP2YOU. For qualitative evaluation, as shown in Fig. 1, UP2YOU exhibits fixed hand skin marks on clothes for all avatars, whereas our reconstruction produces cleaner and more consistent textures across viewpoints.

For system deployment, as shown in Fig. 1, our framework operates in both AR and VR modes. Users wearing an HMD can observe 4DGS avatars driven by the captured motion using GVHMR. Compared with commonly used stylized avatars, such as those from Mixamo, our reconstructed avatars provide higher photorealistic fidelity. Regarding rendering performance, the baseline method outputs a 3D mesh that can be rendered in real time with negligible overhead, but it needs manual rigging to animate the avatar. Our method supports direct animation and achieves approximately 76 FPS, satisfying the real-time requirement for AR/VR applications.

As our long-term goal is to support sports training applications, we note that the influence of different avatar representations on learning performance and user motivation remains underexplored. Future work will conduct user studies to evaluate users’ perception of the visual quality of reconstructed avatars and investigate their influence on the learning process.

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